



R, RStudio, and Quarto

Zhaoxia Yu

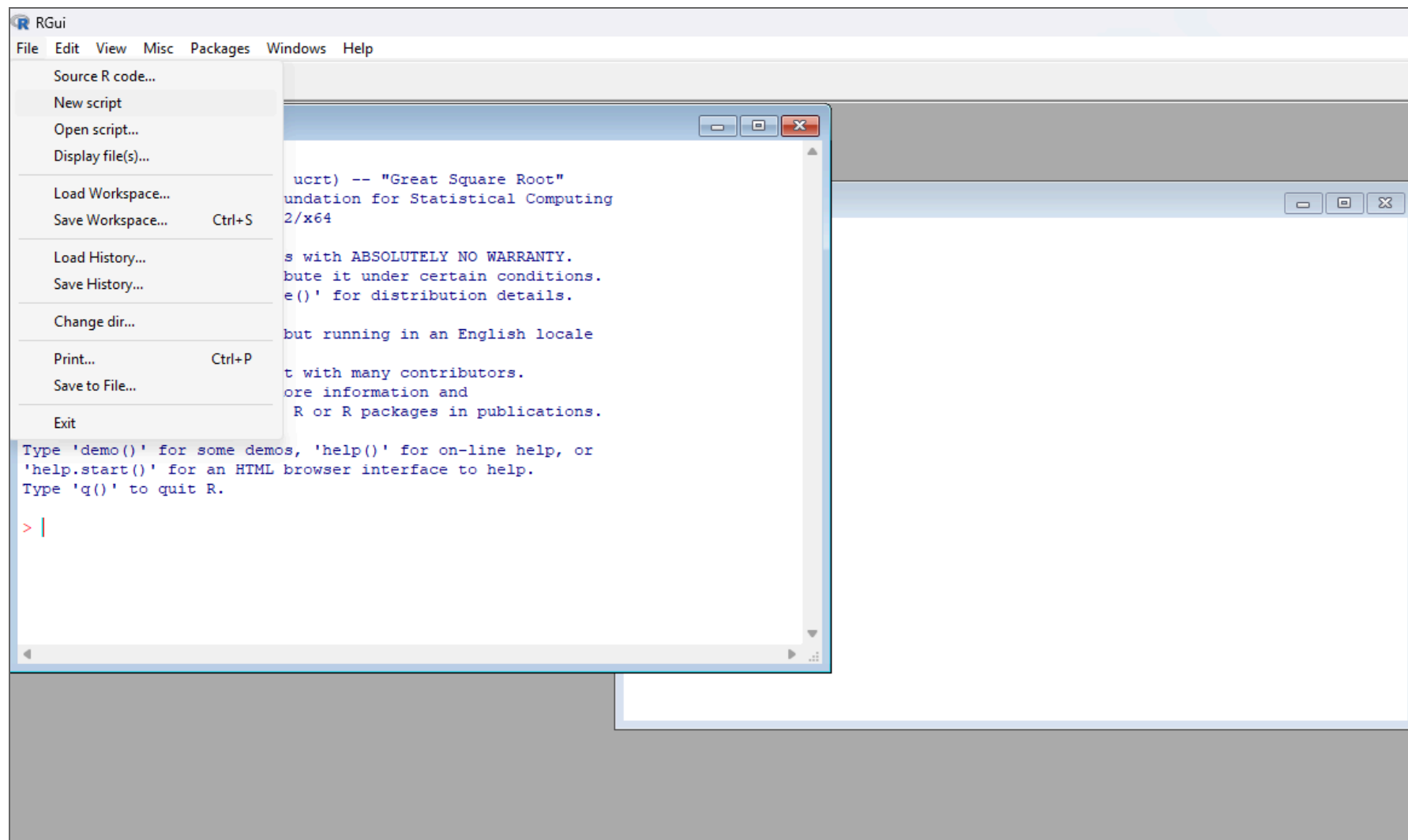
R



What Is R?

- A programming language and software environment for statistical computing and data analysis.
- Created by statisticians Ross Ihaka and Robert Gentleman at the University of Auckland in the early 1990s.
- Runs on Windows, macOS, and Linux.
- Widely used in academia, industry, government, and research around the world.
- Supports data analysis, visualization, machine learning, and reproducible research through thousands of contributed packages.

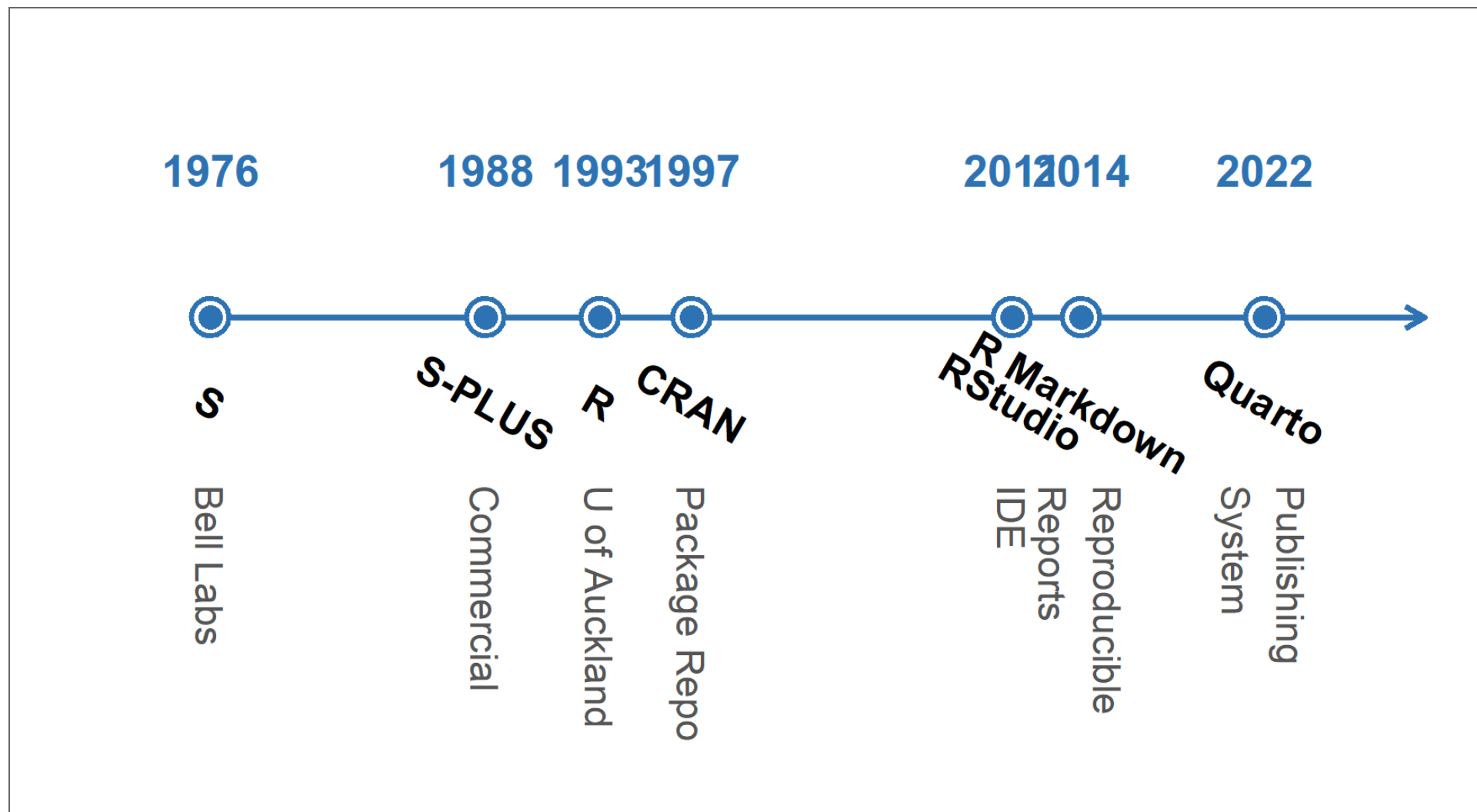
How Does R Look Like?



Why R?

- **Designed for statistics.** Created by statisticians for statistical computing and education.
- **Excellent graphics.** Create publication-quality figures with packages such as **ggplot2**.
- **Large scientific community.**
 - How many packages?
 - **20,000+** packages on CRAN
- **Open source.** Free to use, free to share, and free to modify.
- **Reproducible research.** Analysis, figures, and reports can be generated from a single source.

Evolution of Statistical Computing



R Ecosystem

R is much more than a programming language.

- **R**: The programming language
- **CRAN**: Repository of 20,000+ contributed packages
- **RStudio**: An integrated development environment (IDE) for R
- **RMarkdown / Quarto**: publishing systems for reports, slides, websites, and books
- **Git & GitHub**: Version control and collaboration (optional but highly recommended)

Today we'll use: R + RStudio + Quarto

Getting Started with R

To use R effectively, we need three pieces of software:

1. **R**: the programming language. The engine.
2. **RStudio** (recommended). The dashboard.
 - Where we write and run R code.
3. **Quarto** (optional but highly recommended). The publishing system.
 - Where we create reproducible reports and presentations.

Today's workflow: R → RStudio → Quarto

Installing the Software

Installation order: **R** → **RStudio** → **Quarto**

1. **R**

- The programming language.
- Everything else depends on it. If R is not installed, RStudio will not work.

2. **RStudio**

- Detects and uses your R installation

3. **Quarto** (optional)

- Works with R and RStudio to create reports, slides, and websites

RStudio



What Is RStudio?

- **RStudio** is an integrated development environment (IDE) for R.
- Easy to use: write, run, and debug R code in one place.
- Organized workspace: editor, console, environment, plots, files, and help.
- Productivity tools: syntax highlighting, code completion, project management, and package management.
- Works seamlessly with Quarto to create reports, presentations, and websites.

How Does RStudio Look Like?

The screenshot displays the RStudio environment with a presentation slide titled "Evolution of Statistical Computing". The slide content is as follows:

```

85 - reproducible research. Analysis,
86 figures, and reports can be generated from
87 a single source.
88 ## Evolution of Statistical computing {
89 background-image="img/cosmos-uci-dshs.png"
90 background-size="45px"
91 background-position="20px 95%"}
92 {r}
93 #| echo: false
94 library(ggplot2)
95 library(grid) # for unit()
96
97 timeline <- data.frame(
98   year = c(1976, 1988, 1993, 1997, 2011,
99           2014, 2022),
100   label = c(
101     "S",
102     "S-PLUS",
103     "R",
104     "CRAN",
105     "R Markdown",
106     "RStudio",
107     "Quarto"
108   )
109 )
110
111 ggplot(timeline) +
112   geom_point(aes(year, label)) +
113   geom_text(aes(year, label, size=12)) +
114   theme_minimal()
115
116 
```

The slide also includes a timeline diagram showing the evolution of statistical computing from 1976 to 2022. The timeline is represented by a horizontal line with points corresponding to the years and the associated software or system:

- 1976: S (Bell Labs)
- 1988: S-PLUS (Commercial)
- 1993: R (U of Auckland)
- 1997: CRAN (Package Repo)
- 2011: R Markdown (Reproducible Reports)
- 2014: RStudio (IDE)
- 2022: Quarto (Publishing System)

The RStudio interface shows the source editor on the left, the environment and history panes on the right, and the presentation viewer at the bottom. The presentation viewer displays the slide content and the timeline diagram.

Why RStudio?

- An integrated development environment (IDE) for writing, running, and debugging R code.
- User-friendly interface with separate panes for the script editor, console, environment, plots, and files.
- Improves productivity by providing syntax highlighting, code completion, project management, and package management.
- Works seamlessly with R and Quarto to create reproducible reports, presentations, and websites.
- Free, open source, and running on Windows, macOS, and Linux.

RStudio Ecosystem

RStudio is more than an editor. It integrates with many tools

- Git & GitHub
- Quarto
- Python
- SQL
- Jupyter Notebooks

Today we'll focus on **Quarto**, which allows us to combine text, code, figures, and results into a single document.

From Code to Communication

Writing code is only part of data science. We also need to communicate our work. Quarto allows us to combine

- text
- R code
- tables
- figures

into one reproducible document.

Next, let's learn **Quarto**.

Quarto



What Is Quarto?

- **Quarto** is an open-source scientific and technical publishing system.
- It combines text, code, figures, and results in a single document.
- It works with multiple programming languages, including R, Python, and Julia.
- Quarto is a standalone application that can be used with or without R.
- We will use Quarto together with R and RStudio.

Why Quarto?

- Reproducible research: analysis and results are generated from the same source file.
- Easy to update: changing the data automatically updates tables and figures.
- Professional output: produce reports, presentations, websites, and books.
- Works seamlessly with RStudio.

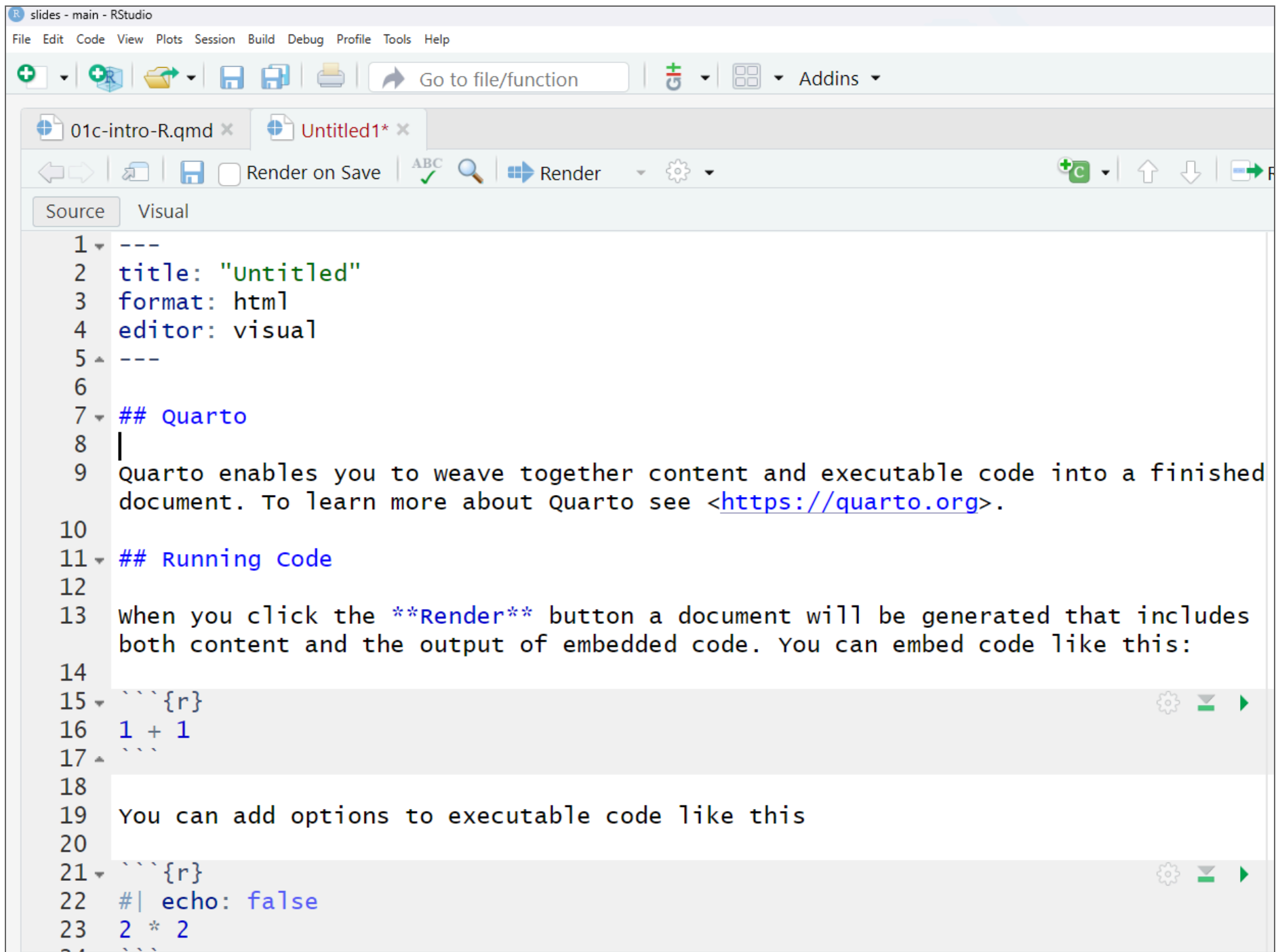
What Can Quarto Create?

Quarto can generate

- PDF reports
- HTML reports
- Reveal.js presentations
- Websites
- Books

These lecture slides were created with Quarto!

Anatomy of a Quarto Document



slides - main - RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function

01c-intro-R.qmd x Untitled1* x

Render on Save

Source Visual

```
1 ---
2 title: "Untitled"
3 format: html
4 editor: visual
5 ---
6
7 ## Quarto
8 |
9 Quarto enables you to weave together content and executable code into a finished
10 document. To learn more about Quarto see <https://quarto.org>.
11 ## Running Code
12
13 when you click the Render button a document will be generated that includes
14 both content and the output of embedded code. You can embed code like this:
15 ```{r}
16 1 + 1
17 ```
18
19 You can add options to executable code like this
20
21 ```{r}
22 #| echo: false
23 2 * 2
24 ```
```


RStudio + Quarto

RStudio makes Quarto easy to use.

- Edit a `.qmd` file.
- Write text using Markdown.
- Add R code.
- Click **Render** to produce the final document.

Summary

Today we introduced three essential tools:

- **R** computes.
- **RStudio** helps us write and organize code.
- **Quarto** communicates results.

Together they form a modern workflow for data science.

Speaker notes